Block practice · Radians, sectors, triangles and 3D trigonometry

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Lengths in a 3D solid using right-angled triangles

Area of a triangle using half ab sin C

Using the cosine rule, including when it gives a quadratic

Using the sine rule, including the two-answer case

Angle between two planes using a right-angled triangle

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Practice workbook · Lengths in a 3D solid using right-angled triangles

Finding diagonals, heights and sloping edges of pyramids, prisms and cuboids by using Pythagoras in one or two right-angled triangles.

22 question parts: 1 worked-example retry and 21 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-13

1. Question 11 · part a · 3 marks

2025 NOV · Paper 02

Worked-example retry · Attempt part a only.

11

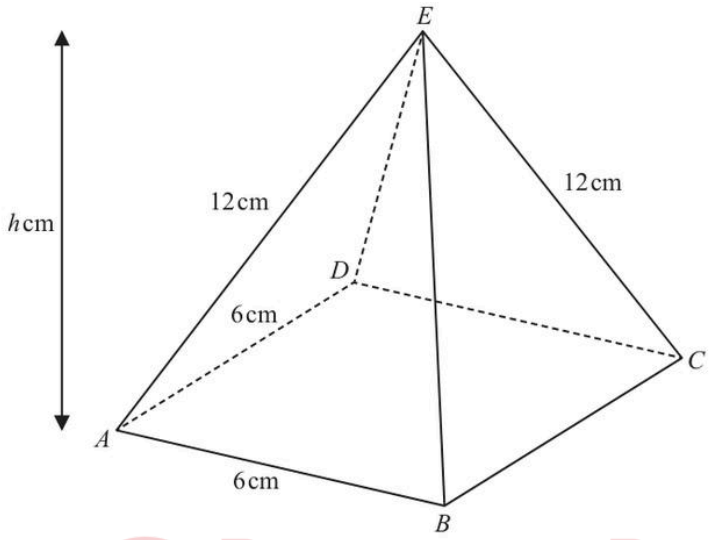


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid $ABCDE$ with

- vertex E
- horizontal square base $ABCD$ of side 6 cm
- $EA = EB = EC = ED = 12$ cm
- vertical height of h cm

(a) Show that $h = 3\sqrt{14}$

(3)

(b) Find, in degrees to one decimal place, the size of the angle between EB and the plane $ABCD$

(2)

The midpoint of AE is P and the midpoint of BE is Q

(c) Find, to the nearest degree, the size of the obtuse angle between the plane EPQ and the plane $PQCD$

(6)

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1. Question 11 · part a · 3 marks · Working and answer

Working and answer

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2. Question 3 · part a · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

3

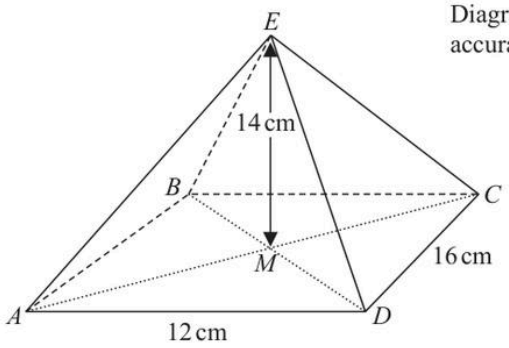


Figure 1

Figure 1 shows the right pyramid $ABCDE$. The base, $ABCD$, of the pyramid is a horizontal rectangle with $AD = 12$ cm and $CD = 16$ cm. The height ME of the pyramid is 14 cm where M is the point of intersection of the diagonals of the base.

The sloping edges, EA, EB, EC and ED of the pyramid are all of equal length.

(a) Calculate, to 3 significant figures, the length of a sloping edge. **(3)**

Calculate, in degrees to one decimal place, the size of

(b) the angle between AE and the base, **(3)**

(c) the angle between the plane AED and the base. **(3)**

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P 6 2 2 8 0 A 0 6 3 2

2. Question 3 · part a · 3 marks · Working and answer

Working and answer

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3. Question 8 · part b · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

8

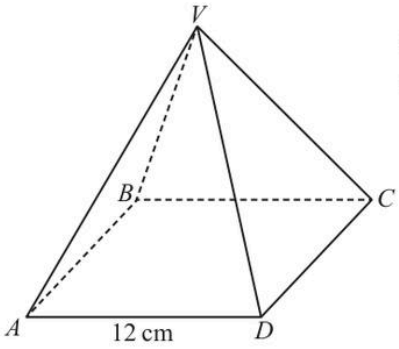


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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3. Question 8 · part b · 3 marks · Working and answer

Working and answer

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4. Question 8 · part b · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

8

Diagram **NOT** accurately drawn

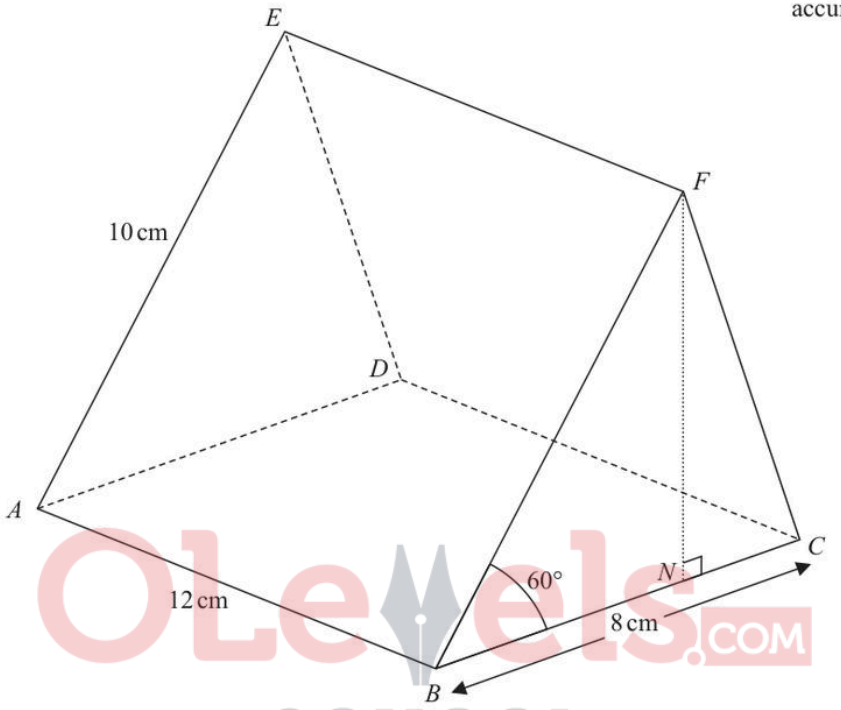


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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P 6 1 8 8 3 A 0 2 2 3 6

4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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5. Question 9 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

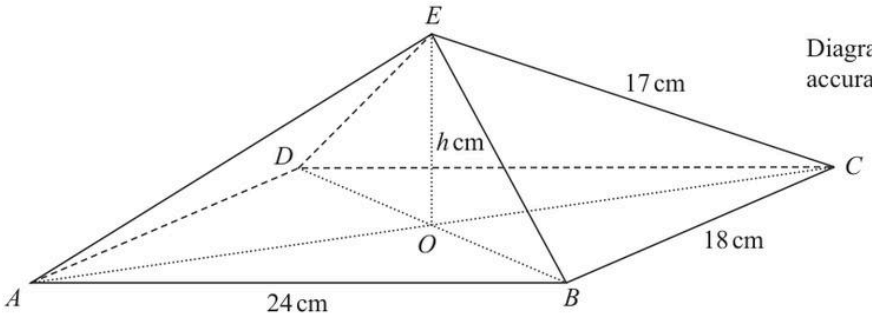


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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P 6 6 3 0 7 A 0 2 2 3 2

5. Question 9 · part b · 3 marks · Working and answer

Working and answer

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6. Question 11 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

11

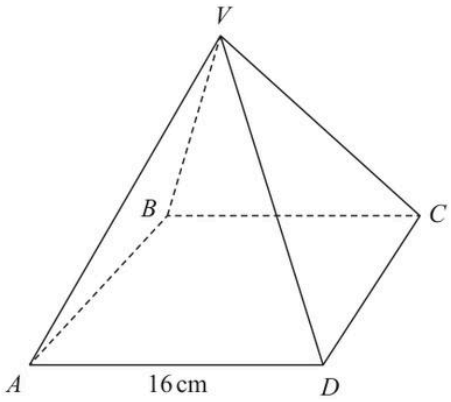


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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6. Question 11 · part b · 3 marks · Working and answer

Working and answer

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7. Question 4 · part a · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part a only.

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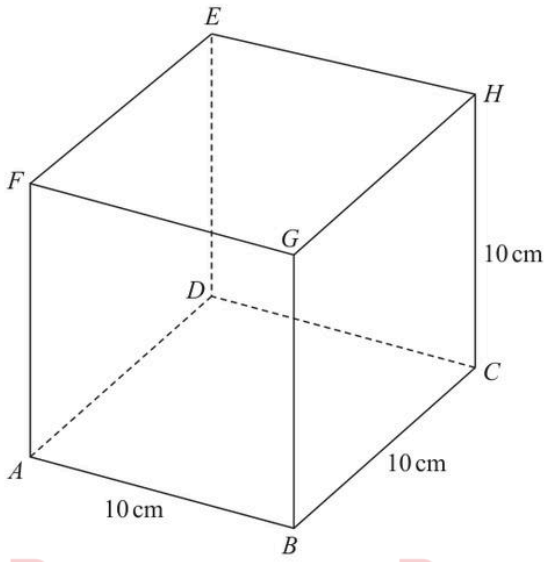


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a solid cube $ABCDEFGH$ with sides of length 10 cm.

(a) Find, in cm, the exact length of FC

(2)

(b) Find, in degrees to one decimal place, the size of $\angle FCA$

(2)

(c) Find, to the nearest degree, the size of the obtuse angle between the plane BFH and the plane BHC

(5)

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8

P 6 6 3 0 8 A 0 8 3 6

7. Question 4 · part a · 2 marks · Working and answer

Working and answer

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8. Question 3 · part a · 2 marks

2020 JAN · Paper 02

Further practice · Attempt part a only.

3

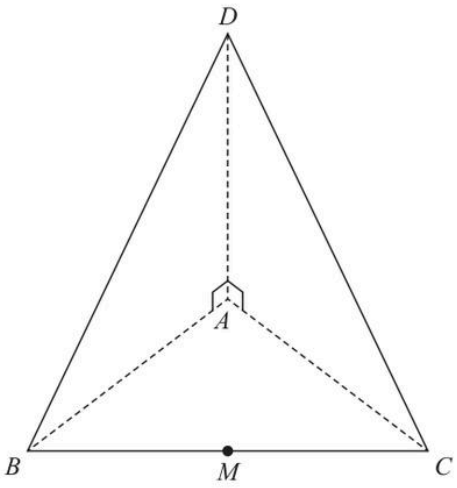


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangular pyramid $ABCD$.

The base, ABC, of the pyramid is a horizontal isosceles triangle with $AB = AC = 10\,\text{cm}$ and $BC = 16\,\text{cm}$. The midpoint of BC is M.

The face BCD of the pyramid is an isosceles triangle with $BD = CD = 26\,\text{cm}$ and D is vertically above A.

$\angle BAD = \angle CAD = 90°$

(a) Calculate the length, in cm, of AM.

(2)

Calculate, in degrees to the nearest degree,

(b) the size of $\angle BCD$,

(3)

(c) the size of the angle between the planes BCA and BCD.

(4)

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P 5 9 9 3 9 A 0 6 3 6

8. Question 3 · part a · 2 marks · Working and answer

Working and answer

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9. Question 9 · part a · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part a only.

9

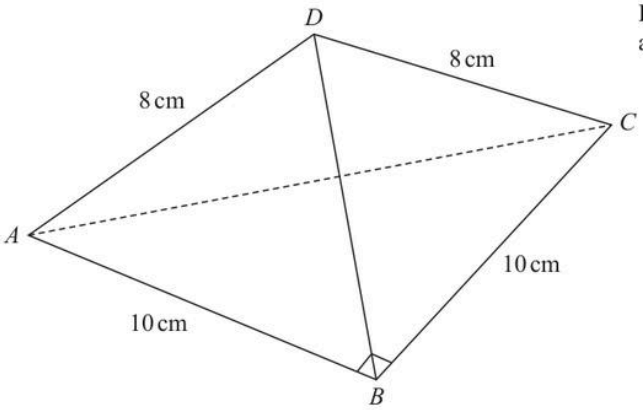


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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26

P 6 6 0 2 7 R R A 0 2 6 3 6

9. Question 9 · part a · 2 marks · Working and answer

Working and answer

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10. Question 9 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

9

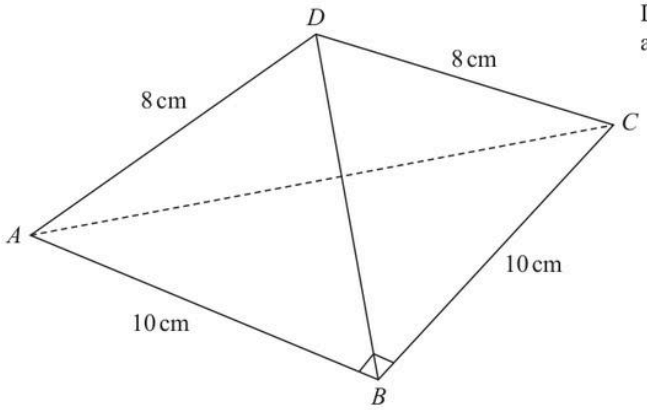


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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10. Question 9 · part b · 2 marks · Working and answer

Working and answer

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11. Question 9 · part a · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part a only.

9

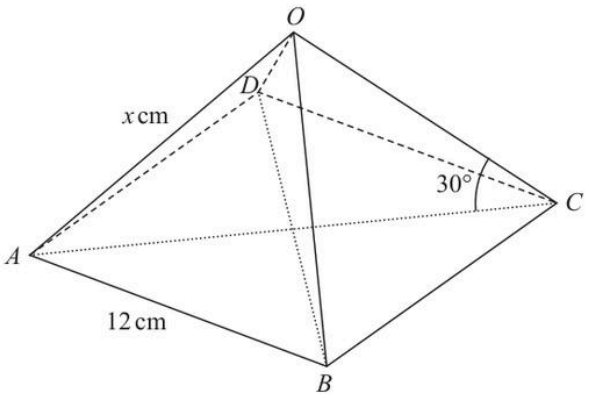


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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P 7 1 8 1 8 A 0 2 2 2 8

11. Question 9 · part a · 2 marks · Working and answer

Working and answer

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12. Question 6 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

6

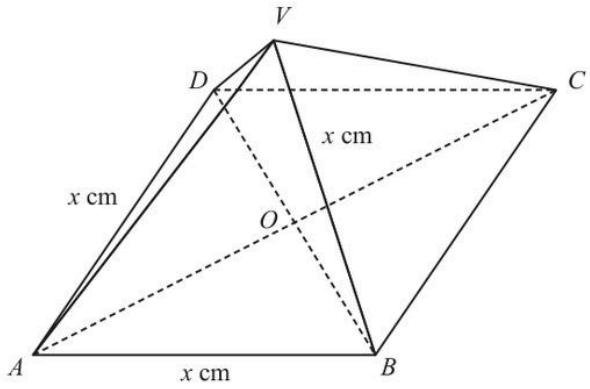


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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P 7 3 5 8 4 A 0 1 6 3 6

12. Question 6 · part a · 3 marks · Working and answer

Working and answer

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13. Question 8 · part a · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part a only.

8

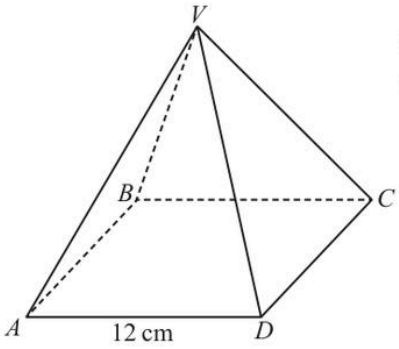


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with vertex V and square base, $ABCD$, of side 12 cm

The size of the angle between the plane VAB and the base $ABCD$ is $30°$

(a) Show that the height of the pyramid is $2\sqrt{3}$ cm

(2)

(b) Find, in cm, the exact length of VA

(3)

(c) Find, in cm^2, the exact area of triangle VAD

(3)

(d) Find, in cm, the exact length of the perpendicular from D to VA

(2)

(e) Find, in degrees to one decimal place, the size of the obtuse angle between the plane VAB and the plane VAD

(3)

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13. Question 8 · part a · 2 marks · Working and answer

Working and answer

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14. Question 8 · part a · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

8

Diagram **NOT** accurately drawn

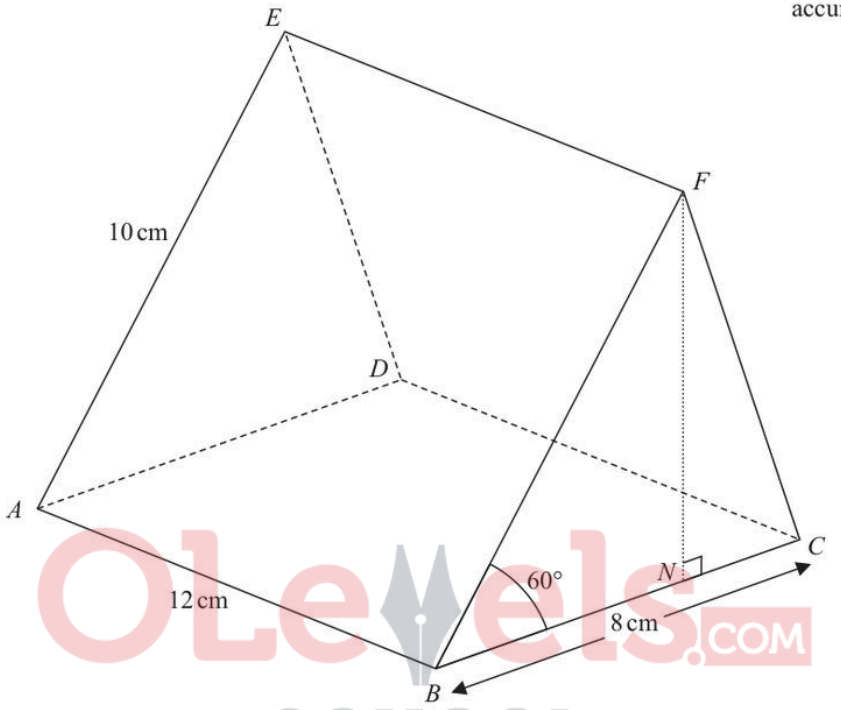


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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P 6 1 8 8 3 A 0 2 2 3 6

14. Question 8 · part a · 2 marks · Working and answer

Working and answer

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15. Question 9 · part a · 1 mark

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

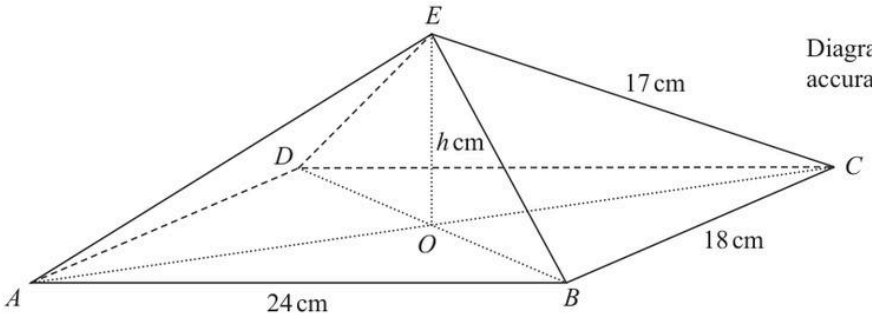


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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P 6 6 3 0 7 A 0 2 2 3 2

15. Question 9 · part a · 1 mark · Working and answer

Working and answer

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16. Question 11 · part a · 4 marks

2019 JUN · Paper 02

Further practice · Attempt part a only.

11

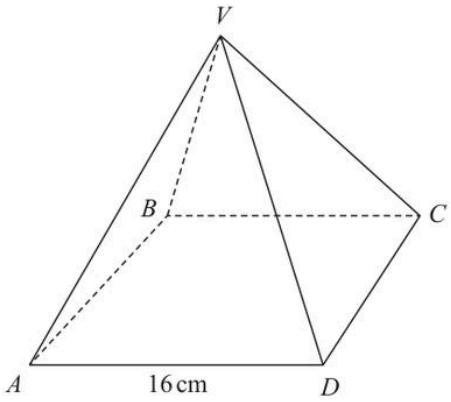


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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16. Question 11 · part a · 4 marks · Working and answer

Working and answer

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17. Question 5 · part a · 3 marks

2023 JAN · Paper 02

Further practice · Attempt part a only.

5

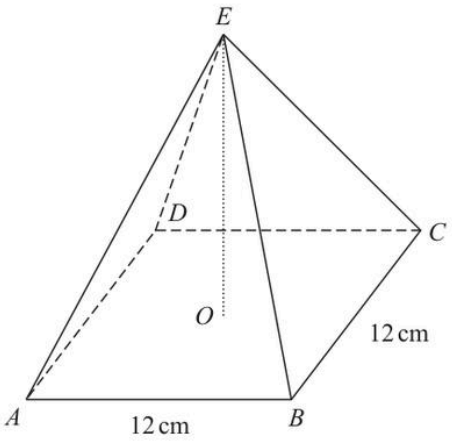


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a right pyramid with a square base $ABCD$ and vertex E.

The base of the pyramid is horizontal with $AB = BC = 12$ cm.
The diagonals of the base intersect at the point O.

The vertex E of the pyramid is vertically above O and the angle between EA and the plane $ABCD$ is 30°

The height of the pyramid is h cm.

(a) Find the exact value of h

(3)

The point F lies on AD such that $AF : FD = 1 : 4$

(b) Calculate, to the nearest degree, the size of the angle EFO.

(4)

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17. Question 5 · part a · 3 marks · Working and answer

Working and answer

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18. Question 10 · part a · 3 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

10

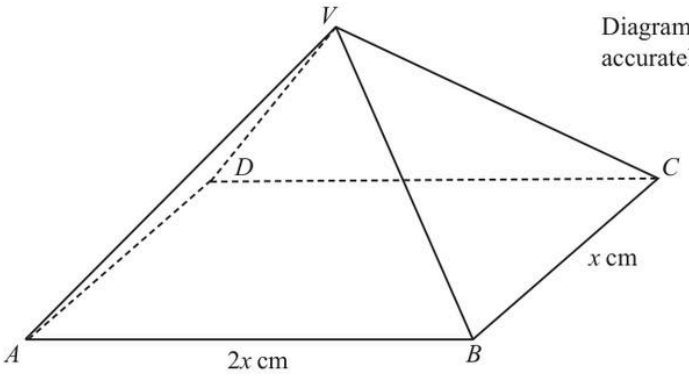


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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18. Question 10 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

19. Question 10 · part b · 2 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

10

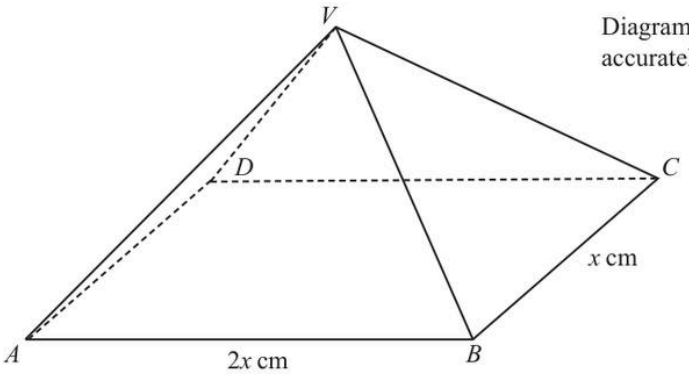


Figure 4

Figure 4 shows a right pyramid $ABCDV$

The base of the pyramid is a rectangle where,

$AB = DC = 2x$ cm $AD = BC = x$ cm

The edges VA, VB, VC and VD are all of equal length.

The angle between VA and $ABCD$ is 45°

(a) Show that $VA = \frac{\sqrt{10}}{2}x$ cm

(3)

(b) Find in cm, the exact height of the pyramid in terms of x

(2)

Find, in degrees to one decimal place,

(c) the size of angle VBA

(2)

(d) the size of the obtuse angle between the plane AVC and the plane BVD

(4)

Given that the volume of the pyramid is $9\sqrt{5}$ cm^3

(e) find the value of x

(2)

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19. Question 10 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

20. Question 9 · part b · 2 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

9

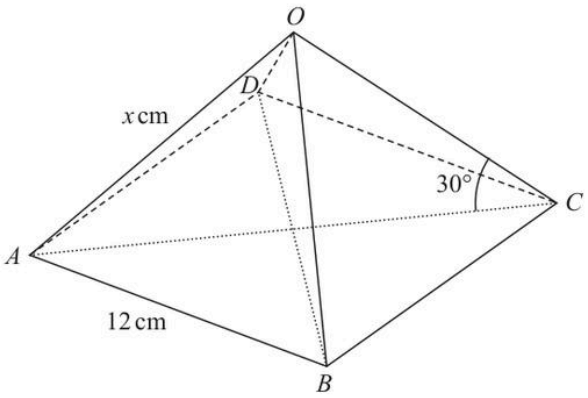


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the right pyramid $OABCD$ with a square base $ABCD$ of side 12 cm.

$OA = OB = OC = OD = x$ cm and $\angle OAC = \angle ODB = \angle OCA = \angle OBD = 30°$

(a) Find the exact length of AC

(2)

(b) Show that $x = 4\sqrt{6}$

(2)

(c) Find the total surface area, to the nearest cm^2, of the pyramid.

(5)

(d) Find the size of the obtuse angle, to the nearest degree, between the plane OAB and the plane OBC

(4)

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P 7 1 8 1 8 A 0 2 2 2 8

20. Question 9 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

9

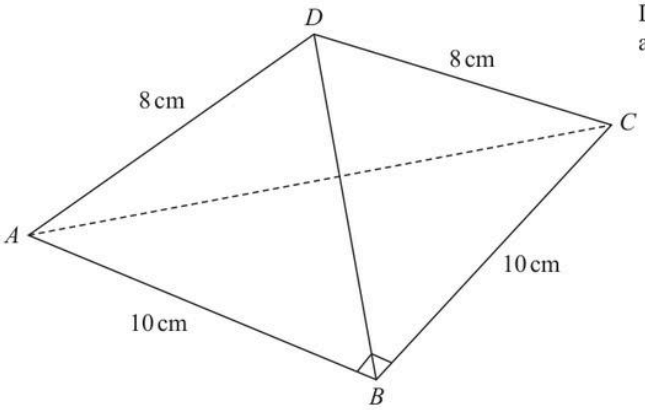


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a triangular pyramid $ABCD$ with base ABC

$AB = BC = 10$ cm $\quad AD = CD = 8$ cm $\quad \angle ABC = 90°$

(a) Find the exact length of AC

Give your answer in the form $p\sqrt{q}$ cm where p is an integer and q is a prime number.

(2)

The point M is the midpoint of AC

(b) Find the exact length of BM

Give your answer in the form $m\sqrt{n}$ cm where both m and n are prime numbers.

(2)

Given that $BD = 6$ cm,

(c) find, in degrees to one decimal place, the size of the acute angle between the plane ACD and the plane ABC

(4)

The base ABC of the pyramid is placed on a horizontal plane.

(d) Find, in cm to 3 significant figures, the vertical height of D above the base.

(2)

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26

P 6 6 0 2 7 R R A 0 2 6 3 6

21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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21. Question 9 · part d · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 10 marks)

21. Question 9 · part d · 2 marks · Working and answer

Working and answer

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22. Question 9 · part c · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part c only.

9

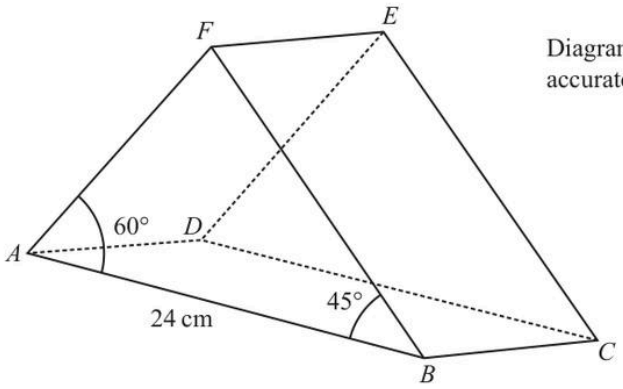


Figure 5

Figure 5 shows a right triangular prism $ABCDEF$ where $ABCD$ is a rectangle.

$AF = DE$ $BF = CE$ $AD = FE = BC$ $AB = DC = 24$ cm

$\angle ABF = \angle DCE = 45°$ $\angle BAF = \angle CDE = 60°$

Using a formula from page 2,

(a) show that $\sin AFB = \dfrac{\sqrt{2} + \sqrt{6}}{4}$

(3)

Without using a calculator,

(b) show that $BF = 12\left(3\sqrt{2} - \sqrt{6}\right)$ cm

(5)

The angle between the plane AEB and the plane $ABCD$ is 65°

(c) Find, in cm to 2 significant figures, the length of EF

(3)

(d) Find, in degrees to one decimal place, the size of the angle between the line CF and the plane $ABCD$

(4)

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22. Question 9 · part c · 3 marks · Working and answer

Working and answer

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Practice workbook · Area of a triangle using half ab sin C

Using the area formula (1/2)ab sin C for a triangle, forwards or backwards, including the case where a given area allows an acute or an obtuse included angle.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-11

1. Question 1 · part whole · 4 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part whole only.

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1 In triangle ABC, $AB = 2x$ cm, $BC = 3x$ cm and $AC = 4x$ cm

The area of triangle ABC is 50 cm^2

Find, to 2 decimal places, the value of x

(4)

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(Total for Question 1 is 4 marks)

1. Question 1 · part whole · 4 marks · Working and answer

Working and answer

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2. Question 2 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

2

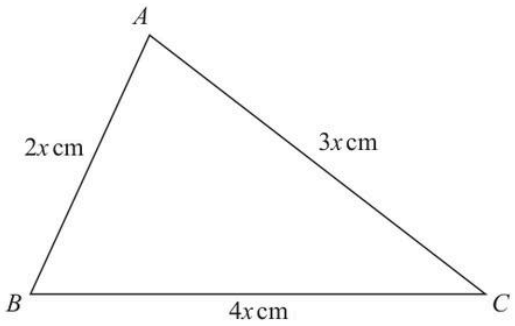


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC in which

$AB = 2x$ cm $\quad AC = 3x$ cm $\quad BC = 4x$ cm

(a) Show that $\sin ABC = \frac{3\sqrt{15}}{16}$

(4)

Given that the area of triangle ABC is $\frac{75\sqrt{15}}{64}$ cm^2

(b) find the value of x.

(2)

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4

2. Question 2 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 6 marks)

2. Question 2 · part b · 2 marks · Working and answer

Working and answer

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3. Question 2 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

2 In triangle ABC, $AB = 3x$ cm, $BC = 5x$ cm and $\angle ABC = 110°$

(a) Find, in degrees to one decimal place, the size of $\angle BCA$

(4)

The area of triangle ABC is 24 cm^2

(b) Find, to 3 significant figures, the value of x

(3)

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3. Question 2 · part b · 3 marks

2023 NOV · Paper 02

Further practice · Attempt part b only.

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Question 2 continued

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(Total for Question 2 is 7 marks)

3. Question 2 · part b · 3 marks · Working and answer

Working and answer

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4. Question 3 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

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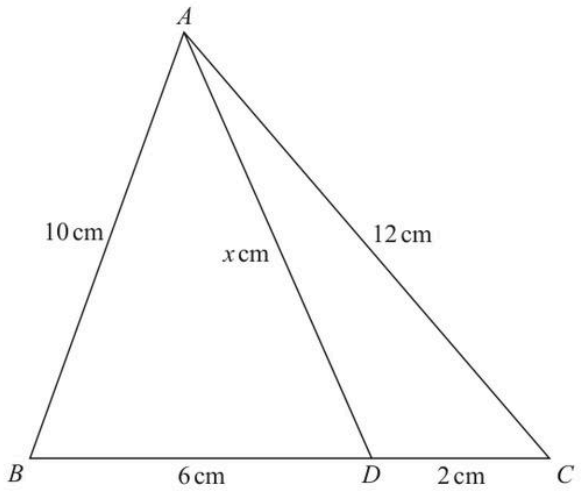


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the triangle ABC in which $AB = 10\,\text{cm}$ and $AC = 12\,\text{cm}$.
The point D lies on BC such that $BD = 6\,\text{cm}$, $DC = 2\,\text{cm}$ and $AD = x\,\text{cm}$.

(a) Show that $x = 11$

(4)

(b) Find the area, in cm^2 to 3 significant figures, of triangle ADB.

(4)

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P 6 2 2 8 4 A 0 6 3 2

4. Question 3 · part b · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.



Question 3 continued

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(Total for Question 3 is 8 marks)

4. Question 3 · part b · 4 marks · Working and answer

Working and answer

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5. Question 3 · part a · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

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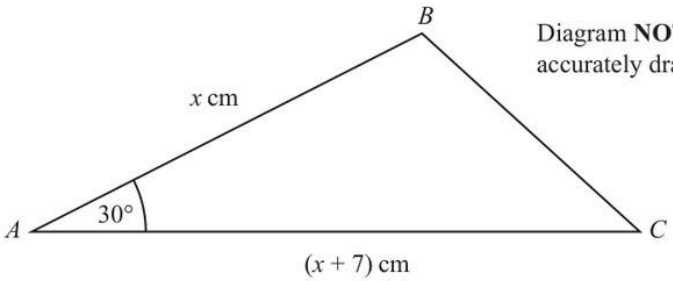


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$ **(3)**

(b) Find, in cm to 3 significant figures, the length of BC **(2)**

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$ **(3)**

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P 7 4 0 9 9 A 0 6 3 2

5. Question 3 · part a · 3 marks · Working and answer

Working and answer

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6. Question 5 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

5 In triangle ABC, $AC = 12$ cm, $BC = 14$ cm, $AB = x$ cm and angle $ABC = 30°$

(a) Show that $x = P\sqrt{3} \pm \sqrt{95}$ where P is a prime number.

(5)

(b) Hence or otherwise, find in cm^2, the difference between the two possible areas of triangle ABC

Give your answer in the form $m\sqrt{n}$ where m is a prime number and n is an integer.

(3)

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6. Question 5 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

Question 5 continued

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(Total for Question 5 is 8 marks)

6. Question 5 · part b · 3 marks · Working and answer

Working and answer

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7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

6 Given that $\dfrac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \dfrac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\dfrac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$

where p, q, r and s are integers.

(3)

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P 7 1 6 4 1 A 0 1 2 3 6

7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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P 7 1 6 4 1 A 0 1 3 3 6

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Turn over

7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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7. Question 6 · part c · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 6 continued

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(Total for Question 6 is 8 marks)

7. Question 6 · part c · 3 marks · Working and answer

Working and answer

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8. Question 3 · part c · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

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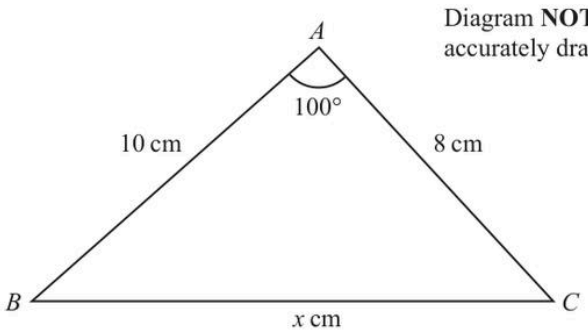


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

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P 7 4 0 9 8 A 0 6 3 2

8. Question 3 · part c · 4 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 9 marks)

8. Question 3 · part c · 4 marks · Working and answer

Working and answer

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9. Question 8 · part b · 4 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

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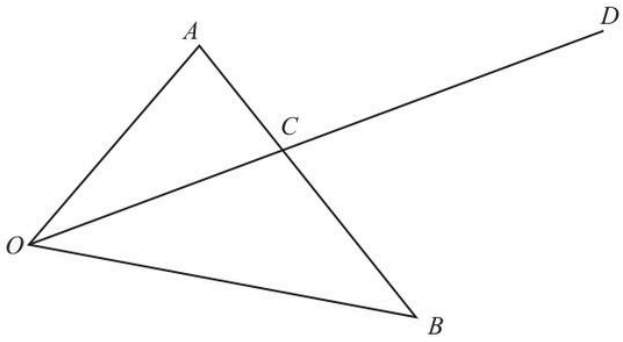


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$

(3)

(b) Find the area of triangle AOB

(4)

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of **a** and **b**

(5)

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9. Question 8 · part b · 4 marks · Working and answer

Working and answer

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10. Question 3 · part b · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

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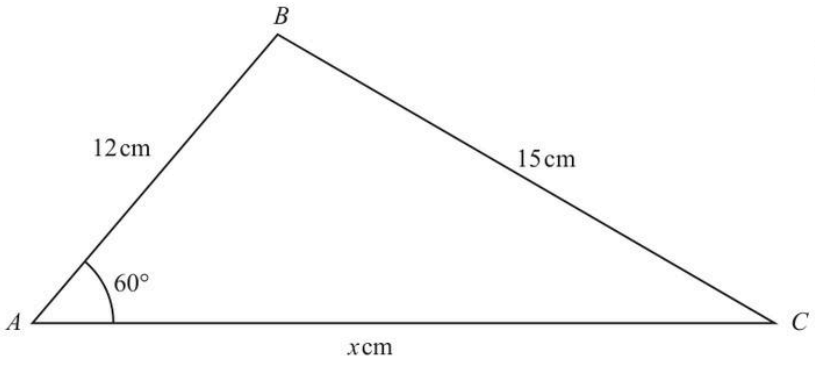


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle ABC where

$AB = 12$ cm $\quad BC = 15$ cm $\quad AC = x$ cm $\quad \angle BAC = 60°$

(a) Show that x satisfies the equation

$$x^2 - 12x - 81 = 0$$

(3)

(b) Hence find the exact area, in cm^2, of triangle ABC

Give your answer in the form $9\left(p\sqrt{q} + \sqrt{r}\right)$ where p, q and r are integers.

(3)

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10. Question 3 · part b · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

Question 3 continued

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(Total for Question 3 is 6 marks)

10. Question 3 · part b · 3 marks · Working and answer

Working and answer

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11. Question 3 · part whole · 5 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

3 In triangle ABC, $AB = 11\,\text{cm}$ and $BC = 12\,\text{cm}$.

The area of triangle $ABC = 33\,\text{cm}^2$

Find, in cm to 3 significant figures, the two possible lengths of AC.

(5)

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11. Question 3 · part whole · 5 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 5 marks)

11. Question 3 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

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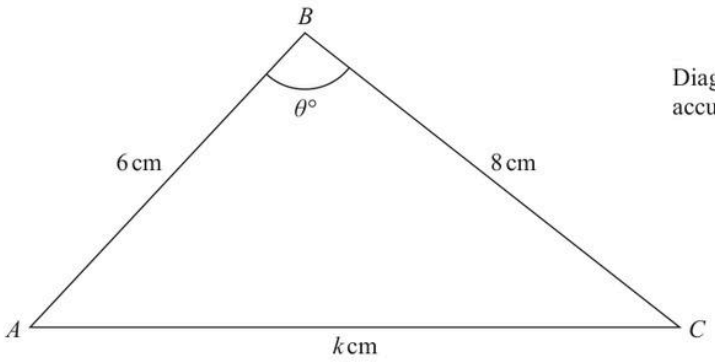


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC

$AB = 6\,\text{cm}$ $BC = 8\,\text{cm}$ $AC = k\,\text{cm}$ $\angle ABC = \theta°$

(a) Show that $\cos\theta° = \dfrac{100 - k^2}{96}$ (2)

The area of triangle ABC is $\sqrt{455}$ cm^2

(b) Find the two possible values of k (7)

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12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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12. Question 7 · part b · 7 marks

2021 NOV · Paper 01

Further practice · Attempt part b only.

Question 7 continued

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(Total for Question 7 is 9 marks)

12. Question 7 · part b · 7 marks · Working and answer

Working and answer

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Practice workbook · Using the cosine rule, including when it gives a quadratic

Using the cosine rule with two sides and the included angle to find the third side, or with three sides to find an angle, including in terms of x.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-07

1. Question 4 · part whole · 5 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.

4 In triangle ABC, $AB = 5x$ cm, $BC = (3x - 1)$ cm, $AC = (2x + 5)$ cm and angle $ABC = 60°$

Find, to 3 significant figures, the value of x.

(5)

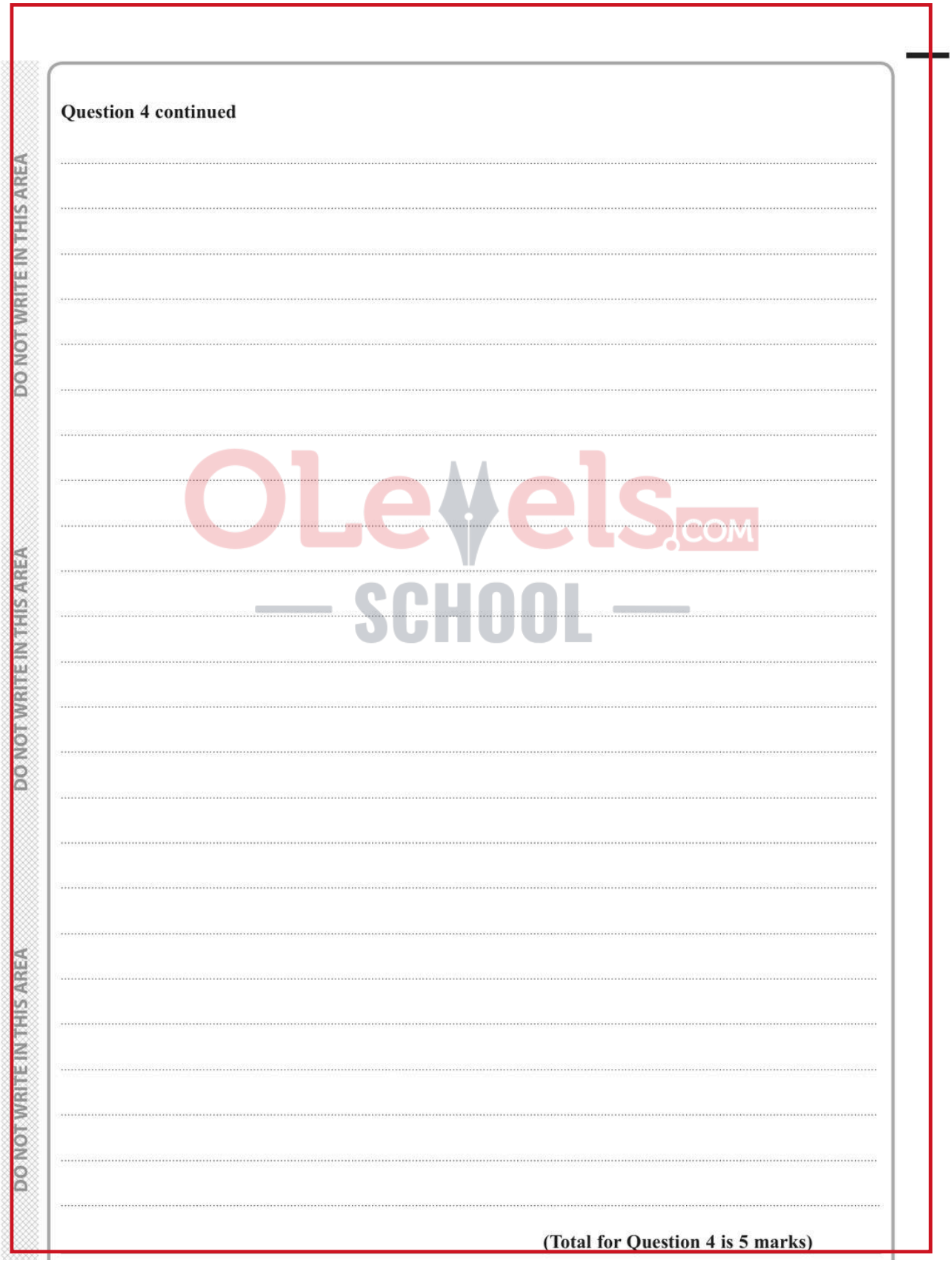
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1. Question 4 · part whole · 5 marks

2019 JUN · Paper 02

Worked-example retry · Attempt part whole only.



Question 4 continued

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(Total for Question 4 is 5 marks)

1. Question 4 · part whole · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

2. Question 3 · part a · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

3

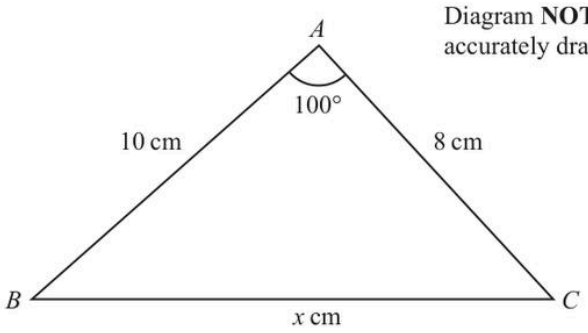


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

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2. Question 3 · part a · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b · 2 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

3

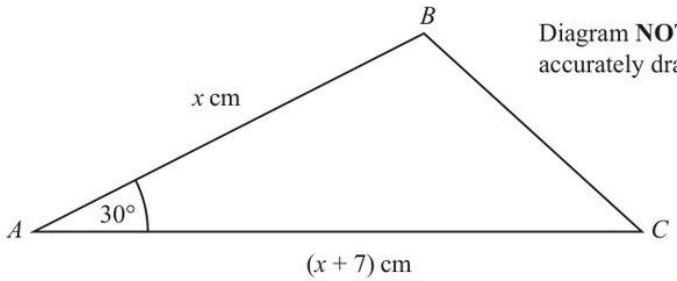


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$

(3)

(b) Find, in cm to 3 significant figures, the length of BC

(2)

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$

(3)

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P 7 4 0 9 9 A 0 6 3 2

3. Question 3 · part b · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

4. Question 8 · part b · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part b only.

8 The point A has coordinates (2, 6), the point B has coordinates (6, 8) and the point C has coordinates (4, 2).

(a) Find the exact length of

(i) AB (ii) BC (iii) AC

(4)

(b) Find the size of each angle of triangle ABC in degrees.

(3)

The points A, B and C lie on a circle with centre P.

(c) Find the coordinates of P.

(2)

(d) Find the exact length of the radius of the circle in the form $\sqrt{a}$, where a is an integer.

(2)

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4. Question 8 · part b · 3 marks · Working and answer

Working and answer

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5. Question 3 · part a · 4 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part a only.

3

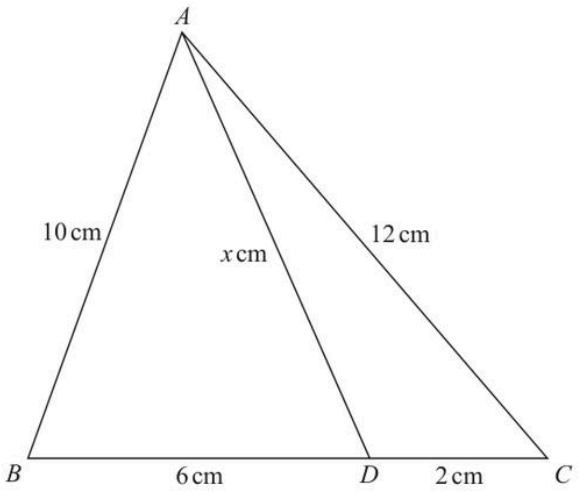


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the triangle ABC in which $AB = 10\,\text{cm}$ and $AC = 12\,\text{cm}$.
The point D lies on BC such that $BD = 6\,\text{cm}$, $DC = 2\,\text{cm}$ and $AD = x\,\text{cm}$.

(a) Show that $x = 11$

(4)

(b) Find the area, in cm^2 to 3 significant figures, of triangle ADB.

(4)

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5. Question 3 · part a · 4 marks · Working and answer

Working and answer

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6. Question 9 · part a · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

9

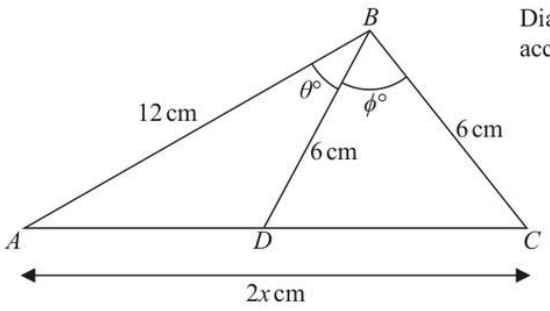


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$

(2)

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$.

(4)

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$

(4)

(d) Hence show that $\theta = 180 - 2\phi$

(2)

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P 6 2 2 8 0 A 0 2 2 3 2

6. Question 9 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

7. Question 7 · part a · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

7

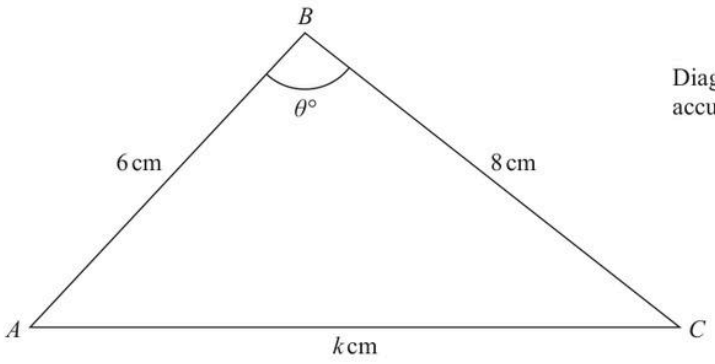


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle ABC

$AB = 6\,\text{cm}$ $BC = 8\,\text{cm}$ $AC = k\,\text{cm}$ $\angle ABC = \theta°$

(a) Show that $\cos\theta° = \dfrac{100 - k^2}{96}$ (2)

The area of triangle ABC is $\sqrt{455}$ cm^2

(b) Find the two possible values of k (7)

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7. Question 7 · part a · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

8. Question 2 · part a · 4 marks

2023 NOV · Paper 02

Further practice · Attempt part a only.

2 In triangle ABC, $AB = 3x$ cm, $BC = 5x$ cm and $\angle ABC = 110°$

(a) Find, in degrees to one decimal place, the size of $\angle BCA$

(4)

The area of triangle ABC is 24 cm^2

(b) Find, to 3 significant figures, the value of x

(3)

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P 7 3 5 8 6 A 0 4 3 2

8. Question 2 · part a · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

9. Question 2 · part whole · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

2

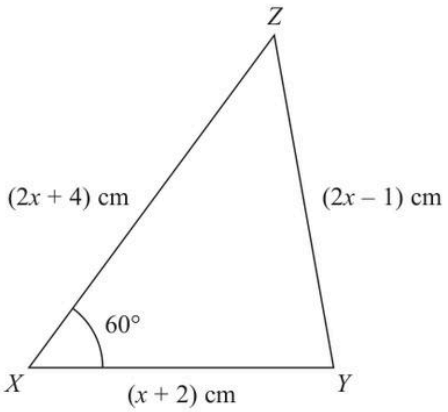


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle XYZ in which

$XY = (x + 2)$ cm $XZ = (2x + 4)$ cm $YZ = (2x - 1)$ cm and $\angle YXZ = 60°$

Find the value of x

Give your answer in the form $p + q\sqrt{3}$ where p and q are integers to be found.

(4)

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9. Question 2 · part whole · 4 marks

2023 JAN · Paper 02

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 4 marks)

9. Question 2 · part whole · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 5 · part a · 5 marks

2025 JUN · Paper 02

Further practice · Attempt part a only.

5 In triangle ABC, $AC = 12$ cm, $BC = 14$ cm, $AB = x$ cm and angle $ABC = 30°$

(a) Show that $x = P\sqrt{3} \pm \sqrt{95}$ where P is a prime number.

(5)

(b) Hence or otherwise, find in cm^2, the difference between the two possible areas of triangle ABC

Give your answer in the form $m\sqrt{n}$ where m is a prime number and n is an integer.

(3)

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10. Question 5 · part a · 5 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 3 · part a · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part a only.

3

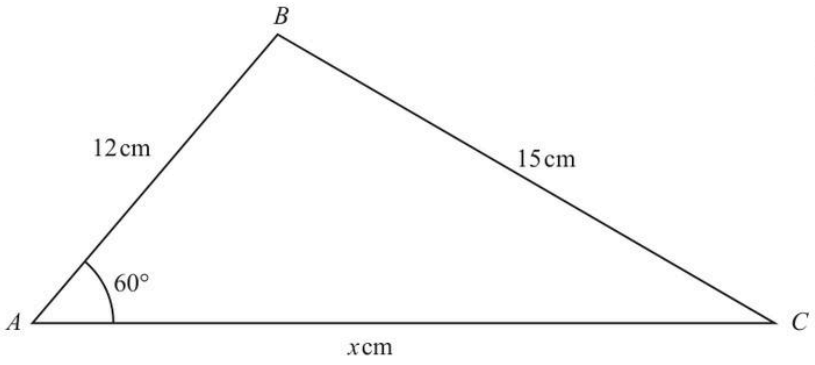


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows triangle ABC where

$AB = 12$ cm $\quad BC = 15$ cm $\quad AC = x$ cm $\quad \angle BAC = 60°$

(a) Show that x satisfies the equation

$$x^2 - 12x - 81 = 0$$

(3)

(b) Hence find the exact area, in cm^2, of triangle ABC

Give your answer in the form $9\left(p\sqrt{q} + \sqrt{r}\right)$ where p, q and r are integers.

(3)

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P 7 8 9 6 4 A 0 6 3 6

11. Question 3 · part a · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

12. Question 9 · part b · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

9

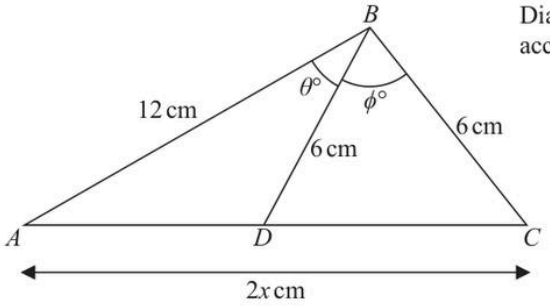


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$

(2)

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$.

(4)

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$

(4)

(d) Hence show that $\theta = 180 - 2\phi$

(2)

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12. Question 9 · part b · 4 marks · Working and answer

Working and answer

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Practice workbook · Using the sine rule, including the two-answer case

Using the sine rule to find a side or an angle, including recognising when an angle can be acute or obtuse.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-09

1. Question 3 · part whole · 5 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part whole only.

3 Triangle ABC is such that

$AC = 10\,\text{cm}$ $BC = 7\text{cm}$ angle $CAB = 25°$

Given that angle ABC is obtuse,

find, in cm to one decimal place, the length of AB

(5)

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1. Question 3 · part whole · 5 marks

2024 NOV · Paper 01

Worked-example retry · Attempt part whole only.

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Question 3 continued

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(Total for Question 3 is 5 marks)

1. Question 3 · part whole · 5 marks · Working and answer

Working and answer

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2. Question 3 · part b.i · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b.i only.

3

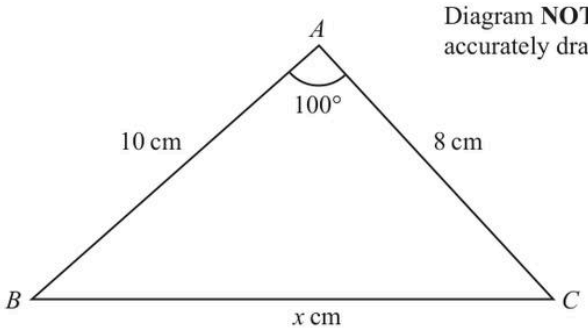


Figure 2

Figure 2 shows triangle ABC where

$$AB = 10 \text{ cm}, \quad AC = 8 \text{ cm}, \quad BC = x \text{ cm} \quad \text{and} \quad \angle BAC = 100^\circ$$

(a) Find, to 3 significant figures, the value of x

(2)

(b) Find, in degrees to one decimal place, the size of

(i) angle ABC (ii) angle ACB

(3)

The bisector of angle ABC meets AC at the point M

(c) Find the area, in cm^2 to 3 significant figures, of triangle BMC.

(4)

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P 7 4 0 9 8 A 0 6 3 2

2. Question 3 · part b.i · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 3 · part c.i · 0 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c.i only.

3

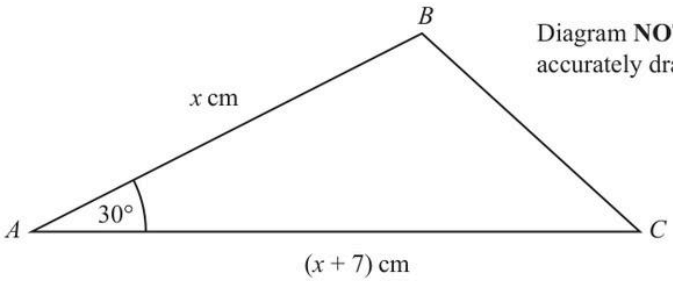


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$ **(3)**

(b) Find, in cm to 3 significant figures, the length of BC **(2)**

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$ **(3)**

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P 7 4 0 9 9 A 0 6 3 2

3. Question 3 · part c.i · 0 marks · Working and answer

Working and answer

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4. Question 3 · part c.ii · 3 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part c.ii only.

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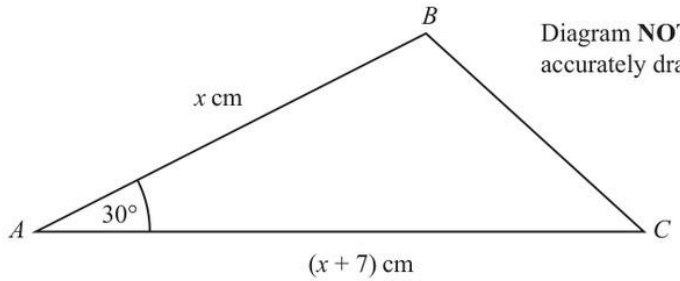


Figure 1

Figure 1 shows triangle ABC where

$AB = x$ cm $\quad AC = (x + 7)$ cm $\quad \angle BAC = 30°$

The area of triangle $ABC = 36$ cm^2

(a) Show that $x = 9$ **(3)**

(b) Find, in cm to 3 significant figures, the length of BC **(2)**

(c) Find, in degrees to one decimal place, the size of

(i) $\angle ABC$

(ii) $\angle ACB$ **(3)**

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4. Question 3 · part c.ii · 3 marks

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Further practice · Attempt part c.ii only.

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Question 3 continued

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(Total for Question 3 is 8 marks)

4. Question 3 · part c.ii · 3 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 2 · part whole · 3 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

2 In ΔABC,

$$\angle BAC = 50° \qquad AB = 10\,\text{cm} \qquad BC = 9\,\text{cm}$$

Given that $\angle BCA = x°$

find the two possible values, to one decimal place, of x

(3)

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5. Question 2 · part whole · 3 marks

2023 JAN · Paper 01 (regional R)

Further practice · Attempt part whole only.

Question 2 continued

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(Total for Question 2 is 3 marks)

5. Question 2 · part whole · 3 marks · Working and answer

Working and answer

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6. Question 4 · part a · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

4 Triangle ABC is such that

$AB = 4.3$ cm $\quad BC = 5.9$ cm $\quad$ angle $BCA = 29°$

(a) Find in degrees, to one decimal place, the two possible values for angle CAB

(4)

(b) Hence, find in cm, to 3 significant figures, the shortest possible length of AC

(3)

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P 7 6 5 0 7 A 0 8 3 2

6. Question 4 · part a · 4 marks · Working and answer

Working and answer

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7. Question 4 · part b · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 Triangle ABC is such that

$AB = 4.3$ cm $\quad BC = 5.9$ cm $\quad$ angle $BCA = 29°$

(a) Find in degrees, to one decimal place, the two possible values for angle CAB

(4)

(b) Hence, find in cm, to 3 significant figures, the shortest possible length of AC

(3)

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P 7 6 5 0 7 A 0 8 3 2

7. Question 4 · part b · 3 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

Question 4 continued

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(Total for Question 4 is 7 marks)

7. Question 4 · part b · 3 marks · Working and answer

Working and answer

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8. Question 3 · part whole · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

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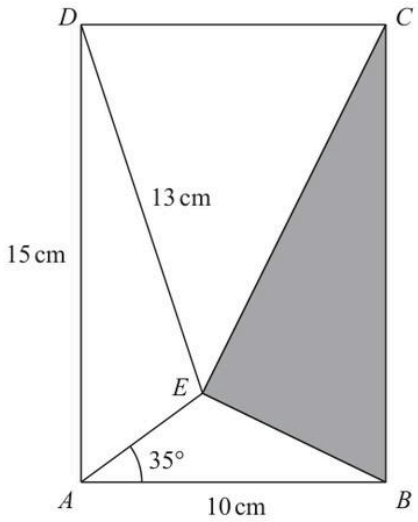


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a rectangle $ABCD$ with $AB = 10$ cm and $AD = 15$ cm.
E is the point inside the rectangle such that $DE = 13$ cm and angle $BAE = 35°$

Given that angle AED is obtuse,

find the area, in cm^2 to one decimal place, of triangle BCE.

(7)

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8. Question 3 · part whole · 7 marks

2022 JUN · Paper 01

Further practice · Attempt part whole only.

Question 3 continued

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(Total for Question 3 is 7 marks)

8. Question 3 · part whole · 7 marks · Working and answer

Working and answer

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9. Question 9 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

9

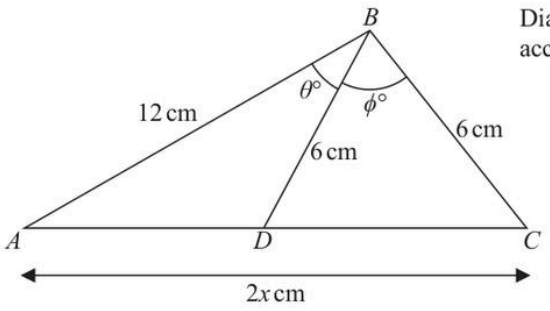


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$

(2)

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$.

(4)

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$

(4)

(d) Hence show that $\theta = 180 - 2\phi$

(2)

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9. Question 9 · part c · 4 marks · Working and answer

Working and answer

OLevels.com SCHOOL

10. Question 9 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

9

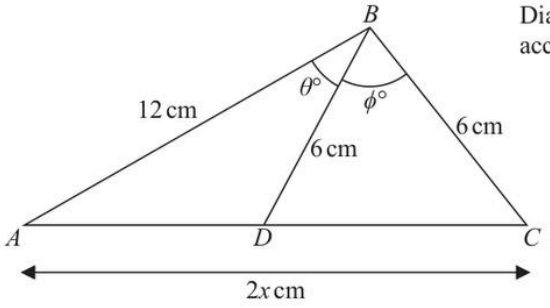


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows triangle ABC with $AB = 12\,\text{cm}$, $BC = 6\,\text{cm}$ and $AC = 2x\,\text{cm}$.

The point D is the midpoint of AC and $BD = 6\,\text{cm}$.

$\angle ABD = \theta°$ and $\angle DBC = \phi°$ where $\theta \neq 0$ and $\phi \neq 0$

(a) Show that $\cos ADB = \dfrac{x^2 - 108}{12x}$ **(2)**

(b) Hence, or otherwise, show that $AC = 6\sqrt{6}\,\text{cm}$. **(4)**

(c) Show that $\sin(\theta° + \phi°) = \sin\phi°$ **(4)**

(d) Hence show that $\theta = 180 - 2\phi$ **(2)**

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10. Question 9 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

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Question 9 continued

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10. Question 9 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.

Question 9 continued

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10. Question 9 · part d · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part d only.



Question 9 continued

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(Total for Question 9 is 12 marks)

10. Question 9 · part d · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

11. Question 6 · part a.i · 0 marks

2022 JAN · Paper 01

Further practice · Attempt part a.i only.

6 In triangle ABC, $AC = x$ cm, $AB = (x + 3)$ cm and $\angle ABC = 30°$

Given that $\angle ACB = \theta°$ where $0 < \theta < 90$

(a) show that

(i) $\sin\theta° = \dfrac{x + 3}{2x}$

(ii) $\cos\theta° = \dfrac{\sqrt{3x^2 - 6x - 9}}{2x}$

(5)

Given that the size of $\angle BAC$: the size of $\angle ABC = 7 : 2$

(b) find the exact value of x

Give your answer in the form $a + a\sqrt{b}$ where a and b are prime numbers.

(5)

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11. Question 6 · part a.i · 0 marks · Working and answer

Working and answer

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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

6 In triangle ABC, $AC = x$ cm, $AB = (x + 3)$ cm and $\angle ABC = 30°$

Given that $\angle ACB = \theta°$ where $0 < \theta < 90$

(a) show that

(i) $\sin\theta° = \dfrac{x + 3}{2x}$

(ii) $\cos\theta° = \dfrac{\sqrt{3x^2 - 6x - 9}}{2x}$

(5)

Given that the size of $\angle BAC$: the size of $\angle ABC = 7 : 2$

(b) find the exact value of x

Give your answer in the form $a + a\sqrt{b}$ where a and b are prime numbers.

(5)

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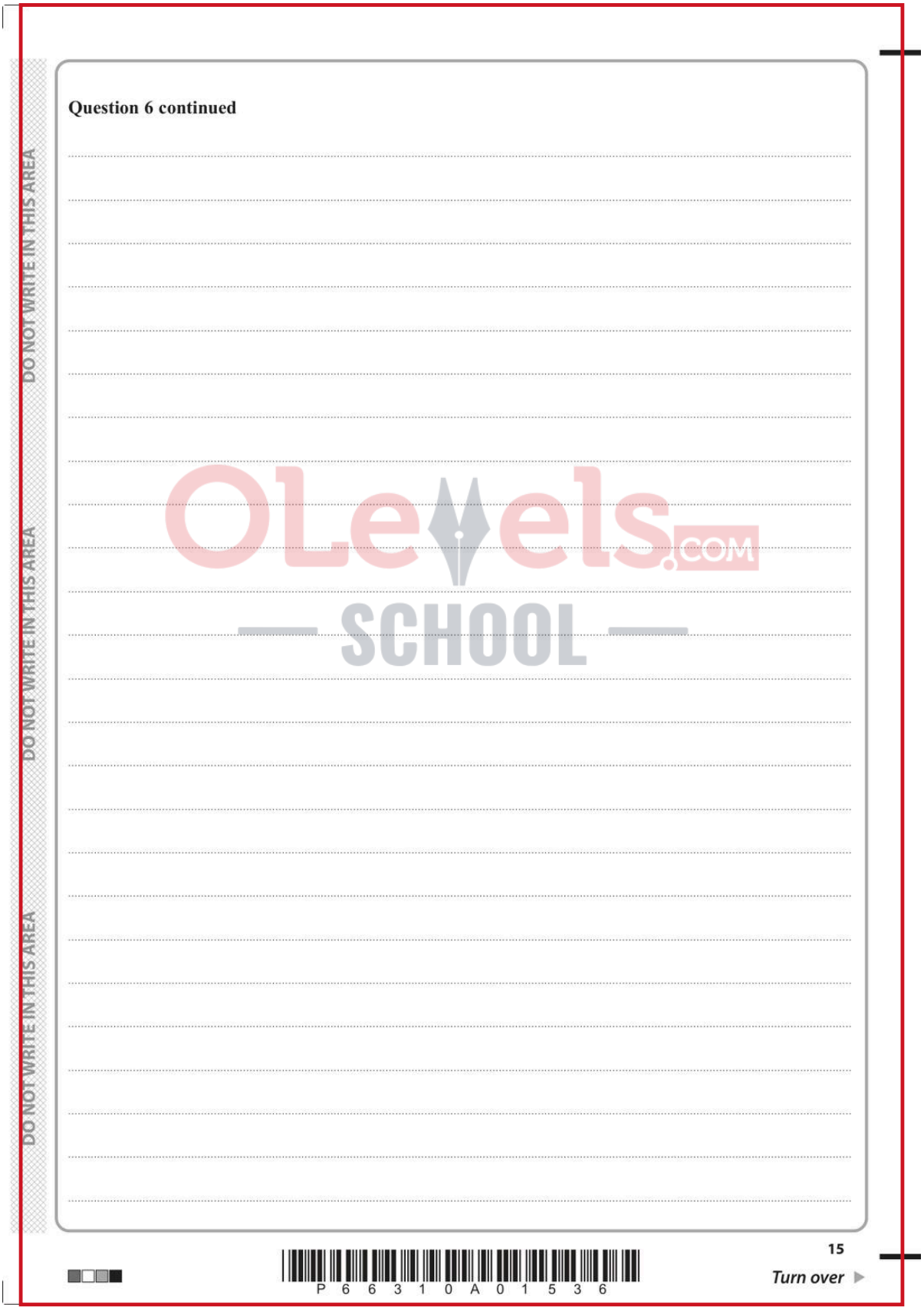
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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.



Question 6 continued

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15

Turn over

12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 6 continued

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12. Question 6 · part b · 5 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

Question 6 continued

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(Total for Question 6 is 10 marks)

12. Question 6 · part b · 5 marks · Working and answer

Working and answer

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13. Question 6 · part b · 3 marks

2022 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 Given that $\frac{a+\sqrt{5}}{\sqrt{5}-2} = 11 + 5\sqrt{5}$

(a) without using a calculator, find the value of a
Show your working clearly.

(2)

Triangle PQR is such that

$PR = (x+3)\,\text{cm}$ $QR = x\,\text{cm}$ angle $QPR = 30°$ angle $PQR = 45°$

(b) Show that $x = 3 + 3\sqrt{2}$

(3)

Given that $\sin 105° = \frac{\sqrt{6}+\sqrt{2}}{4}$ and that the area of triangle PQR is $A\,\text{cm}^2$

(c) find the exact value of A in the form $\frac{9}{8}\left(p\sqrt{6} + q\sqrt{2} + r\sqrt{3} + s\right)$
where p, q, r and s are integers.

(3)

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13. Question 6 · part b · 3 marks · Working and answer

Working and answer

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14. Question 9 · part b · 5 marks

2024 JUN · Paper 02

Further practice · Attempt part b only.

9

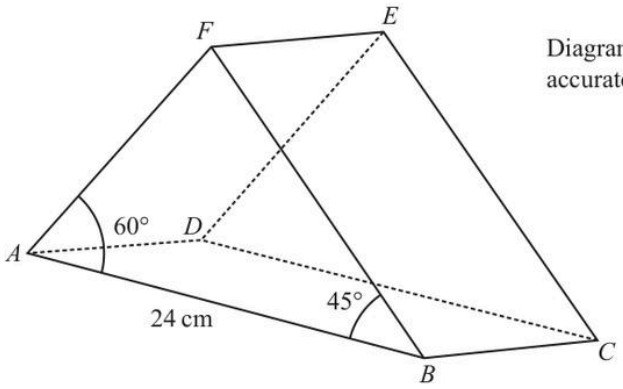


Figure 5

Figure 5 shows a right triangular prism $ABCDEF$ where $ABCD$ is a rectangle.

$AF = DE$ $BF = CE$ $AD = FE = BC$ $AB = DC = 24$ cm

$\angle ABF = \angle DCE = 45°$ $\angle BAF = \angle CDE = 60°$

Using a formula from page 2,

(a) show that $\sin AFB = \dfrac{\sqrt{2} + \sqrt{6}}{4}$

(3)

Without using a calculator,

(b) show that $BF = 12\left(3\sqrt{2} - \sqrt{6}\right)$ cm

(5)

The angle between the plane AEB and the plane $ABCD$ is 65°

(c) Find, in cm to 2 significant figures, the length of EF

(3)

(d) Find, in degrees to one decimal place, the size of the angle between the line CF and the plane $ABCD$

(4)

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14. Question 9 · part b · 5 marks · Working and answer

Working and answer

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Practice workbook · Angle between two planes using a right-angled triangle

Finding the angle between two planes, where one is usually the base, by taking perpendiculars to their common line and working in a right-angled triangle.

11 question parts: 1 worked-example retry and 10 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-TRI-16

1. Question 9 · part whole · 6 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

9

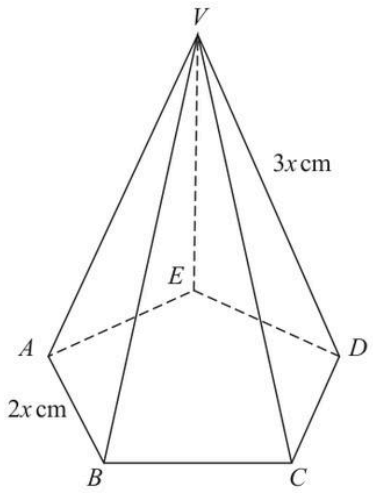


Diagram **NOT** accurately drawn

Figure 4

Figure 4 shows a right pyramid with vertex V and base $ABCDE$ which is a regular pentagon.

$$AB = BC = CD = DE = EA = 2x \text{ cm}$$
$$VA = VB = VC = VD = VE = 3x \text{ cm}$$

Find, in degrees to one decimal place, the size of the angle between the plane VBC and the base $ABCDE$

(6)

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1. Question 9 · part whole · 6 marks

2023 JUN · Paper 02

Worked-example retry · Attempt part whole only.

Question 9 continued

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(Total for Question 9 is 6 marks)

1. Question 9 · part whole · 6 marks · Working and answer

Working and answer

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2. Question 3 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

3

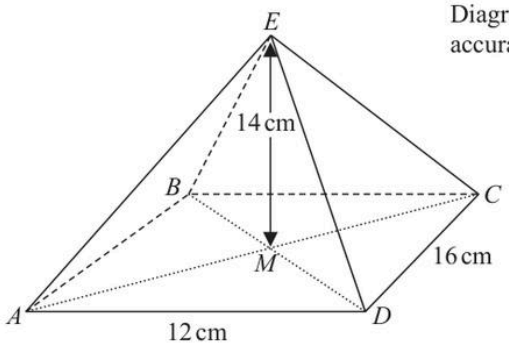


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows the right pyramid $ABCDE$. The base, $ABCD$, of the pyramid is a horizontal rectangle with $AD = 12$ cm and $CD = 16$ cm. The height ME of the pyramid is 14 cm where M is the point of intersection of the diagonals of the base.

The sloping edges, EA, EB, EC and ED of the pyramid are all of equal length.

(a) Calculate, to 3 significant figures, the length of a sloping edge. **(3)**

Calculate, in degrees to one decimal place, the size of

(b) the angle between AE and the base, **(3)**

(c) the angle between the plane AED and the base. **(3)**

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2. Question 3 · part c · 3 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.



Question 3 continued

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(Total for Question 3 is 9 marks)

2. Question 3 · part c · 3 marks · Working and answer

Working and answer

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3. Question 8 · part c · 2 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part c only.

8

Diagram **NOT** accurately drawn

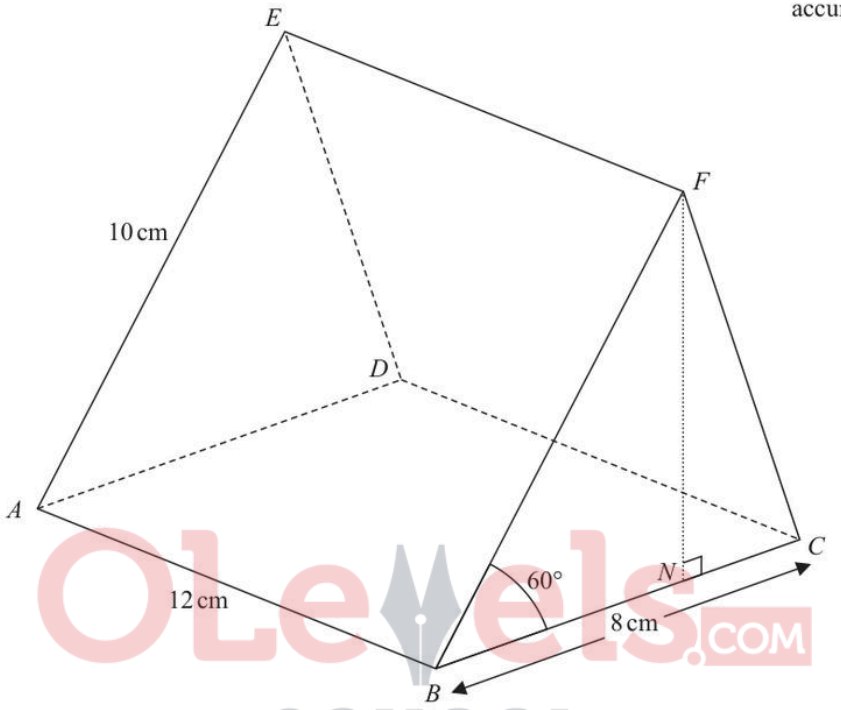


Figure 3

Figure 3 shows a right prism $ABCDEF$. The cross section BCF of the prism is a triangle.

$AB = DC = 12\,\text{cm}$ $BC = AD = 8\,\text{cm}$ $BF = AE = 10\,\text{cm}$ $\angle FBC = \angle EAD = 60°$

The point N lies on BC such that FN is perpendicular to BC.

(a) Show that $BN = 5\,\text{cm}$.

(2)

(b) Find, in cm to 3 significant figures, the length of EN.

(3)

The midpoint of BF is X and the midpoint of FC is Y.

(c) Find, in degrees to one decimal place, the size of the angle between the plane $ABCD$ and the plane $AXYD$.

(2)

(d) Find, in degrees to one decimal place, the size of the angle AYE.

(6)

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3. Question 8 · part c · 2 marks · Working and answer

Working and answer

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4. Question 9 · part c · 2 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

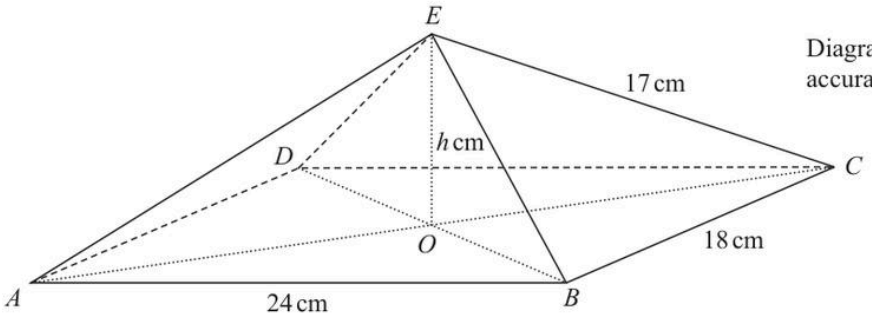


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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4. Question 9 · part c · 2 marks · Working and answer

Working and answer

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5. Question 9 · part d · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

9 Given that α is the acute angle such that $\tan\alpha = \frac{2}{3}$

(a) find the exact value of $\cos\alpha$

(1)

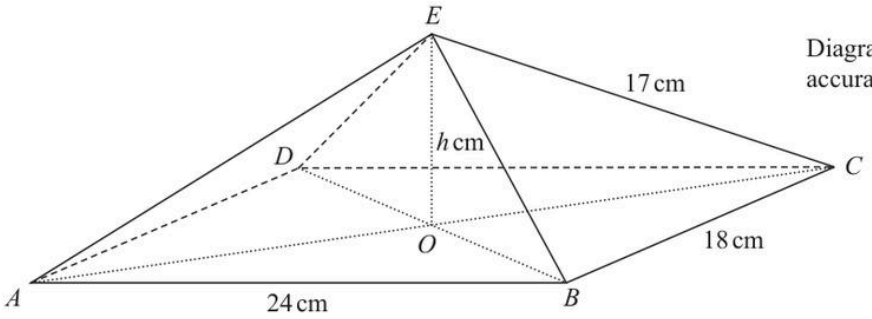


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with a rectangular base $ABCD$ and vertex E

The rectangular base of the pyramid is horizontal with $AB = 24$ cm and $BC = 18$ cm.

The diagonals of the base intersect at the point O

The vertex E of the pyramid is vertically above O such that

$$AE = BE = CE = DE = 17 \text{ cm}$$

The height of the pyramid is h cm.

(b) Find the value of h

(3)

The size of the angle between the plane EBC and the plane $ABCD$ is $\theta°$

(c) Show that $\tan\theta° = \frac{2}{3}$

(2)

The point P is the midpoint of EB and the point Q is the midpoint of EC

(d) Find the size, in degrees to one decimal place, of the angle between the plane OPQ and the plane $BCQP$

(4)

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5. Question 9 · part d · 4 marks

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Further practice · Attempt part d only.

Question 9 continued

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5. Question 9 · part d · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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5. Question 9 · part d · 4 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part d only.

Question 9 continued

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(Total for Question 9 is 10 marks)

5. Question 9 · part d · 4 marks · Working and answer

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6. Question 6 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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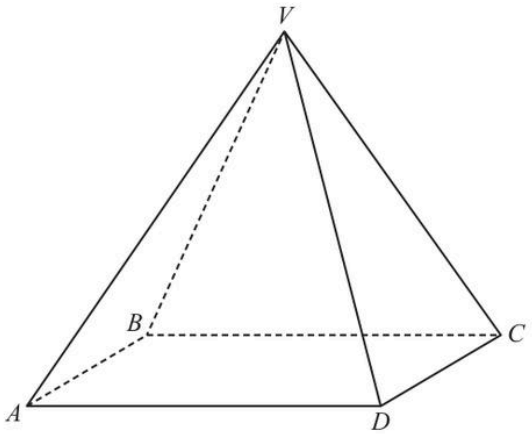


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows a right pyramid $VABCD$ with vertex V and rectangular base $ABCD$

$$AD = VC = 3AB$$

Find in degrees, to one decimal place, the size of the angle between the plane CVD and the base $ABCD$

(6)

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P 7 6 5 0 9 A 0 1 4 3 2

6. Question 6 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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Question 6 continued

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6. Question 6 · part whole · 6 marks

2025 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 6 continued

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6. Question 6 · part whole · 6 marks

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Further practice · Attempt part whole only.

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Question 6 continued

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(Total for Question 6 is 6 marks)

6. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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7. Question 11 · part d · 3 marks

2019 JUN · Paper 02

Further practice · Attempt part d only.

11

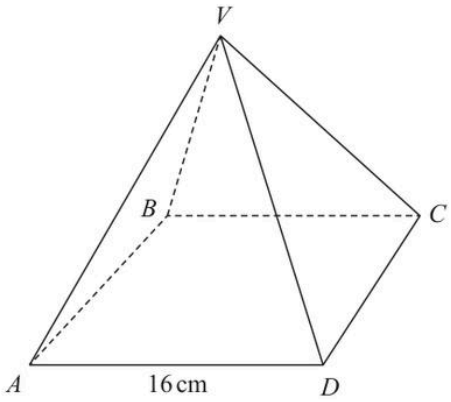


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid with vertex V and square base, $ABCD$, of side 16 cm.

The size of angle AVC is 90°

(a) Show that the height of the pyramid is $8\sqrt{2}$ cm.

(4)

(b) Find, in cm, the length of VA.

(3)

(c) Find, in cm, the exact length of the perpendicular from D onto VA.

(3)

Find, in degrees to one decimal place, the size of

(d) the angle between the plane VAB and the base $ABCD$,

(3)

(e) the obtuse angle between the plane VAB and the plane VAD.

(3)

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7. Question 11 · part d · 3 marks · Working and answer

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8. Question 4 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

4

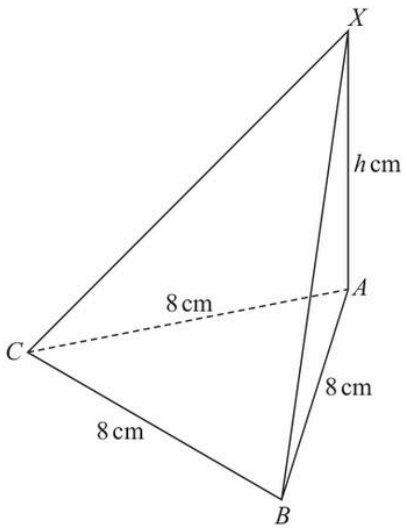


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a triangular pyramid $ABCX$.
The base ABC of the pyramid is an equilateral triangle where $AB = BC = CA = 8$ cm.
The vertex X of the pyramid is such that AX is perpendicular to the base of the pyramid and $AX = h$ cm.

The volume of the pyramid is $48\sqrt{3}$ cm^3

(a) Show that $h = 9$

(3)

(b) Find, in degrees to one decimal place, the size of angle BXC.

(3)

(c) Find, in degrees to one decimal place, the size of the angle between the plane BCX and the base ABC of the pyramid.

(3)

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8. Question 4 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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8. Question 4 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.

Question 4 continued

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8. Question 4 · part c · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part c only.



8. Question 4 · part c · 3 marks · Working and answer

Working and answer

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9. Question 6 · part whole · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

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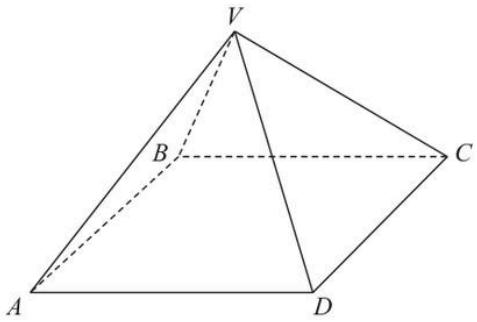


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a right pyramid $VABCD$ with vertex V and square base $ABCD$.

Each of the edges of the pyramid has the same length.

Find the size, in degrees to one decimal place, of the angle between the plane CVD and the base $ABCD$.

(6)

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9. Question 6 · part whole · 6 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part whole only.

Question 6 continued

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(Total for Question 6 is 6 marks)

9. Question 6 · part whole · 6 marks · Working and answer

Working and answer

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10. Question 3 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

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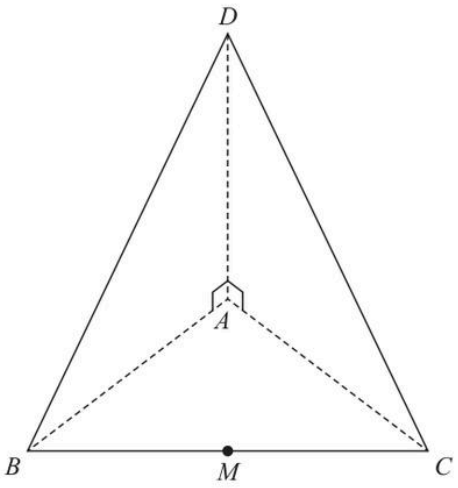


Diagram **NOT** accurately drawn

Figure 1

Figure 1 shows a triangular pyramid $ABCD$.

The base, ABC, of the pyramid is a horizontal isosceles triangle with $AB = AC = 10\,\text{cm}$ and $BC = 16\,\text{cm}$. The midpoint of BC is M.

The face BCD of the pyramid is an isosceles triangle with $BD = CD = 26\,\text{cm}$ and D is vertically above A.

$\angle BAD = \angle CAD = 90°$

(a) Calculate the length, in cm, of AM.

(2)

Calculate, in degrees to the nearest degree,

(b) the size of $\angle BCD$,

(3)

(c) the size of the angle between the planes BCA and BCD.

(4)

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10. Question 3 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 3 continued

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10. Question 3 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 3 continued

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10. Question 3 · part c · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part c only.

Question 3 continued

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(Total for Question 3 is 9 marks)

10. Question 3 · part c · 4 marks · Working and answer

Working and answer

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11. Question 6 · part c · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part c only.

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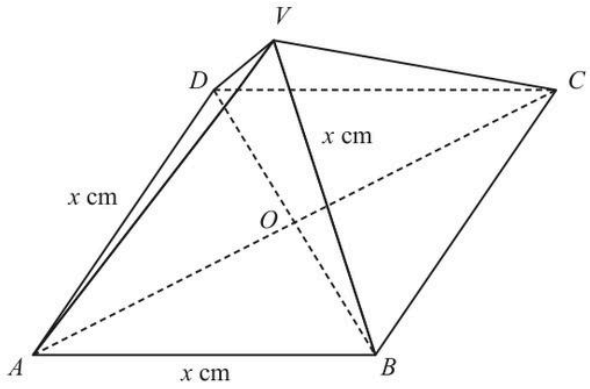


Diagram **NOT** accurately drawn

Figure 3

Figure 3 shows a right pyramid with a horizontal square base.

$$AB = BC = CD = DA = x \text{ cm}$$

$$AV = BV = CV = DV = x \text{ cm}$$

O is the point of intersection of the diagonals of the base.
The vertex V of the pyramid is vertically above O

(a) Show that $VO = \frac{\sqrt{2}}{2}x$ cm

(3)

(b) Find, in degrees, the size of the angle AVC

(2)

(c) Find, in degrees to one decimal place, the size of the angle between the plane VAB and the plane VDC

(3)

The volume of the pyramid is 200 cm^3

Given that the volume of a pyramid $= \frac{1}{3} \times$ base area $\times$ height

(d) Find to 3 signficant figures, the value of x

(3)

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11. Question 6 · part c · 3 marks · Working and answer

Working and answer

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